

CME396 – PROCESS PLANNING AND COST ESTIMATION

CIAT – II (October 2026) • SET A • Answers / Study Material

Units 3, 4 and 5 • III Year / V Semester • Mechanical Engineering

PART A – 2 MARK ANSWERS

1. Distinguish between estimation and costing.

Estimation	Costing
Probable cost worked out before production.	Actual cost worked out during / after production.
Done by the estimator (technical person) from drawings.	Done by the cost accountant from actual records.
Used to fix the selling price and submit quotations.	Used to control cost and find the actual profit.

2. List the elements of cost.

- **Material** – direct material and indirect material.
- **Labour** – direct labour and indirect labour.
- **Expenses** – direct expenses and indirect expenses (overheads: factory, administrative, selling & distribution).

3. A concern produces 600 machines per year. Total overheads = Rs. 1,80,000. Find overhead per machine by unit rate method.

$$\text{Overhead per machine} = \frac{\text{Total overheads}}{\text{Number of units}} = \frac{1,80,000}{600} = \text{Rs. 300 per machine}$$

4. Define depreciation.

Depreciation is the **permanent decrease in the value of an asset** (machine, building, tool) with time, due to wear and tear, age and obsolescence. It is charged as a cost every year so that money is available to replace the asset at the end of its life.

5. What is flash loss in forging shop?

In closed-die (drop) forging, extra metal is provided so that the die cavity fills completely. This excess metal squeezes out between the two die halves along the parting line as a thin fin called **flash**, which is trimmed off later. The metal lost this way is the flash loss. It depends on the **perimeter of the forging × flash width × flash thickness**.

6. Classify the different types of welding joints.

(i) Butt joint (square, single-V, double-V, U, J), (ii) Lap joint, (iii) Tee (T) joint, (iv) Corner joint, (v) Edge joint.

7. Define pattern allowances in foundry.

Pattern allowances are the extra amounts by which a pattern is made larger or different from the final casting so that the casting comes out to the correct size. Types: **shrinkage allowance, machining (finish) allowance, draft (taper) allowance, distortion allowance and shake (rapping) allowance**.

8. State the common cost components in estimating jobs in a welding shop.

- Cost of electrodes / filler rod (and flux).
- Cost of power (arc welding) or gases (oxygen, acetylene).
- Labour cost of welder and helper, including edge preparation and cleaning.
- Overheads (depreciation of welding set, supervision, fixtures).

9. What is tear-down time?

Tear-down time is the time taken, after a batch is finished, to remove the jigs, fixtures and tools from the machine, return them to the store and clean the machine. Like set-up time, it is shared equally by all the pieces of the batch.

10. Write the formula to calculate the time taken for drilling operation in a lathe.

$$N = \frac{1000 V}{\pi d} \text{ rpm} \quad T = \frac{l + 0.3 d}{f \times N} \text{ min}$$

where l = depth of hole (mm), d = drill diameter (mm), $0.3d$ = allowance for the drill point (approach), f = feed (mm/rev), V = cutting speed (m/min).

PART B – 13 MARK ANSWERS

11 (a) Calculate prime cost, factory cost, production cost, total cost and selling price per item.

(Data: opening stock Rs. 25,000; purchases Rs. 40,000; closing stock Rs. 15,000; direct labour Rs. 14,000; direct expenses Rs. 1,000; factory overhead Rs. 9,750; administrative Rs. 6,500; selling & distribution Rs. 3,250; 650 items; profit 10% of total cost.)

Formulae (ladder of cost)

Material consumed = Opening stock + Purchases – Closing stock

Prime cost = Direct material + Direct labour + Direct expenses

Factory cost = Prime cost + Factory overheads

Production cost = Factory cost + Administrative overheads

Total cost = Production cost + Selling & distribution overheads

Selling price = Total cost + Profit

Solution

Step	Working	Total (Rs.)	Per item (Rs.)
Material consumed	25,000 + 40,000 – 15,000	50,000	76.92
Prime cost	50,000 + 14,000 + 1,000	65,000	100.00
Factory cost	65,000 + 9,750	74,750	115.00
Production cost	74,750 + 6,500	81,250	125.00
Total cost	81,250 + 3,250	84,500	130.00
Profit (10% of total cost)	0.10 × 130	8,450	13.00
Selling price	130 + 13	92,950	143.00

Prime cost = Rs. 100, Factory cost = Rs. 115, Production cost = Rs. 125, Total cost = Rs. 130, Selling price = Rs. 143 per item

11 (b) Describe overheads and explain the various methods of allocating overhead charges with suitable examples.

Definition

Overheads are all **indirect costs** – indirect material + indirect labour + indirect expenses – which cannot be charged directly to a particular job. They must be shared out to all the jobs on a fair basis.

Classification

Type	Examples
Factory (works) overhead	Factory rent, power, lighting, repairs, depreciation of machines, supervisor's salary
Administrative (office) overhead	Salaries of office staff, office rent, stationery, telephone, audit fees

Selling overhead	Advertisement, salesmen's salary and commission, showroom expenses
Distribution overhead	Packing, warehouse rent, transport and delivery vans

Methods of allocation

1. Percentage of direct material cost – used when material is the main cost.

$$\% \text{ Overhead} = \frac{\text{Total overheads}}{\text{Total direct material cost}} \times 100$$

2. Percentage of direct labour cost – simple; used when wage rates are uniform.

$$\% \text{ Overhead} = \frac{\text{Total overheads}}{\text{Total direct labour cost}} \times 100$$

3. Percentage of prime cost – considers both material and labour.

$$\% \text{ Overhead} = \frac{\text{Total overheads}}{\text{Total prime cost}} \times 100$$

4. Man-hour rate – time based; suitable where work is manual.

$$\text{Man-hour rate} = \frac{\text{Total overheads}}{\text{Total man-hours}}$$

5. Machine-hour rate – most accurate for machine shops.

$$\text{Machine-hour rate} = \frac{\text{Total machine running cost}}{\text{Total machine hours}}$$

6. Combination of man-hour and machine-hour rates – where both men and machines are important.

7. Unit rate method – when all products are identical: Overhead per unit = Total overheads ÷ Number of units.

Example

Total factory overhead = Rs. 50,000 and total direct labour = Rs. 1,00,000 → overhead = 50% of labour. A job with direct labour of Rs. 2,000 is charged Rs. 1,000 overhead.

Conclusion: machine-hour rate is the most accurate in modern machine shops, since most overheads are caused by machines.

12 (a) Illustrate the step by step procedure for doing cost estimation for a new product of your own.

Definition

Estimation is the technical process of finding the probable cost of a product before production starts, from its drawings and specifications.

Procedure

- 1. Study the drawing and specifications** – material, dimensions, tolerances, surface finish, quantity.
- 2. Prepare the bill of materials** – all parts, bought-out items and quantities.
- 3. Estimate material cost** – volume → weight (volume × density) → add allowances → × rate per kg.

4. **Decide the sequence of operations** and the machine for each.
5. **Estimate time for each operation** – set-up + machining + handling + allowances.
6. **Estimate labour cost** = time × wage rate per hour.
7. **Add direct expenses** – special tools, jigs, patterns, royalties.
8. **Find prime cost** = material + labour + direct expenses.
9. **Add overheads** – factory → factory cost; administrative → production cost; selling & distribution → total cost.
10. **Add profit** – total cost + profit = selling price.

Example – cast iron V-belt pulley.

Item	Working	Rs.
Material	5 kg × Rs. 80 per kg	400
Labour	2 h × Rs. 150 per h	300
Direct expenses	Share of pattern cost	50
Prime cost	400 + 300 + 50	750
Factory overhead	40% of labour	120
Administrative + selling overhead	60 + 70	130
Total cost	750 + 120 + 130	1,000
Selling price	1,000 + 15% profit	1,150

12 (b) (i) **Reducing balance method:** C = Rs. 25,000, n = 20 years, S = Rs. 4,000. (7) (ii) **Machine-hour basis:** shaper C = Rs. 85,000, S = Rs. 8,500, n = 8 years, 3330 h/year. (6)

(i) Reducing balance method

$$r = 1 - \left(\frac{S}{C}\right)^{1/n} = 1 - \left(\frac{4,000}{25,000}\right)^{1/20} = 1 - (0.16)^{0.05} = 1 - 0.9124 = 0.0876 = \mathbf{8.76\%}$$

[log 0.16 = -0.7959; × 0.05 = -0.0398; antilog = 0.9124]

Year	Value at start (Rs.)	Depreciation @ 8.76% (Rs.)	Value at end (Rs.)
1	25,000.00	2,188.90	22,811.10
2	22,811.10	1,997.24	20,813.86
3	20,813.86	1,822.33	18,991.53

Depreciation fund after 3 years = 25,000 – 18,991.53 = Rs. 6,008.47

Depreciation for 3rd year = Rs. 1,822.33; value reduced by 8.76% every year

(ii) Machine-hour basis

Total working hours = 8 × 3,330 = 26,640 h

$$\text{Depreciation per hour} = \frac{C - S}{\text{Total hours}} = \frac{85,000 - 8,500}{26,640} = \mathbf{\text{Rs. 2.87 per hour}}$$

Annual depreciation = 2.8716 × 3,330 = **Rs. 9,562.50 per year** (11.25% of cost)

13 (a) Explain the various types of losses incurred in the estimation of forging shop with examples.

Introduction

In forging, the weight of stock to be cut from the bar (gross weight) is always more than the weight of the finished forging (net weight), because some metal is lost in the process.

$$\text{Gross weight} = \text{Net weight} + \text{Total losses}$$

Types of losses

1. **Shear (cutting) loss** – when long bars are cut into small pieces, the end pieces of the bar are too short to be used. Taken as about 5% of net weight.
2. **Scale loss** – on heating, the surface oxidises and forms iron oxide scale, which falls off during hammering. Taken as about **6% of net weight** for the first heating plus about 1% for every further heating.
3. **Flash loss** – in closed-die forging, excess metal squeezes out at the parting line and is trimmed off.

$$\text{Flash volume} = \text{Perimeter of forging} \times \text{Flash width} \times \text{Flash thickness}$$

4. **Tonghold loss** – a length of bar is held by the tongs during forging and is cut off later. Usually 1 to 1.5 times the bar diameter in length.

$$\text{Tonghold volume} = \frac{\pi}{4} d^2 \times (1 \text{ to } 1.5 d)$$

5. **Sprue loss** – in drop forging, the connection between the forging and the tonghold (sprue) is cut off; about 7% of net weight.
6. **Burr loss (punching loss)** – when holes are punched, the metal removed (burr) is lost. Burr volume = area of hole $\times$ thickness.

Example

A forging has net weight 2.0 kg. Losses: scale 6% = 0.12 kg; flash 4% = 0.08 kg; tonghold 0.10 kg; shear 5% = 0.10 kg.

Gross weight = 2.0 + 0.12 + 0.08 + 0.10 + 0.10 = **2.40 kg**. At Rs. 80 per kg, material cost = Rs. 192.

Conclusion: ignoring these losses under-estimates the material cost by 15–25%, so they must always be added.

13 (b) Discuss the procedure for production cost estimation of a welding shop with a suitable case example for modern welding.

Elements of welding cost

- Cost of electrode / filler material
- Cost of power (arc) or gas (gas welding)
- Labour cost (welding + edge preparation + cleaning)
- Overheads (depreciation of welding set, supervision, fixtures)

Procedure

1. **Study the joint** – type of joint, plate thickness, length of weld, edge preparation.
2. **Cross-section of weld** – e.g. single-V butt: $A = g t + t^2 \tan(\theta/2)$; fillet weld: $A = \frac{1}{2} s^2$.

3. **Weight of weld metal** = $A \times L \times \text{density}$.

4. **Electrode required** = Weight of weld metal $\div$ deposition efficiency (allows for stub-end, spatter and slag losses).

5. **Electrode cost** = Electrode weight $\times$ Rate per kg.

6. **Arc time** = Length of weld $\div$ welding speed; **total time** = Arc time $\div$ operator factor.

7. **Labour cost** = Total time $\times$ Labour rate.

8. **Power cost** = $\frac{V \times I \times \text{Arc time}}{1000 \times \eta} \times \text{Rate per kWh}$.

9. **Overheads** added as % of labour; **Total welding cost** = Electrode + Power + Labour + Overheads.

Case example – MIG / arc welding of a fillet joint

Fillet weld of 6 mm leg, 500 mm long, density 7.8 g/cm³, deposition efficiency 80%, wire Rs. 150/kg, speed 0.25 m/min, operator factor 60%, labour Rs. 180/h, overhead 100% of labour, 25 V, 180 A, $\eta = 70\%$, power Rs. 8/kWh.

- $A = \frac{1}{2} \times 6^2 = 18 \text{ mm}^2 \rightarrow \text{volume} = 18 \times 500 = 9,000 \text{ mm}^3 = 9 \text{ cm}^3 \rightarrow \text{weld metal} = 9 \times 7.8 = 70.2 \text{ g}$
- Wire required = $70.2 \div 0.8 = 87.75 \text{ g} \rightarrow \text{cost} = 0.08775 \times 150 = \text{Rs. } 13.16$
- Arc time = $0.5 \div 0.25 = 2 \text{ min}$; total time = $2 \div 0.6 = 3.33 \text{ min} \rightarrow \text{labour} = (3.33/60) \times 180 = \text{Rs. } 10.00$; overhead = Rs. 10.00
- Power = $(25 \times 180 \div 0.7) \text{ W} \times (2/60) \text{ h} = 0.214 \text{ kWh} \rightarrow \text{Rs. } 1.71$

Total welding cost $\approx 13.16 + 10.00 + 10.00 + 1.71 = \text{Rs. } 34.87$

In modern welding (MIG/TIG, robotic), deposition efficiency and operator factor are higher, so labour cost per metre falls while equipment overhead rises.

14 (a) Steel chain of 60 m length; link of 6 mm bar, two straight portions of 30 mm and semicircular ends of 15 mm inside radius. Find the total length and weight of bar. Density = 7.7 g/cm³.

Solution

Diameter of bar $d = 6 \text{ mm}$, so the mean (centre-line) radius of the ends = $15 + 3 = 18 \text{ mm}$.

$$\text{Length of bar per link} = 2 \times 30 + 2 \times \pi \times 18 = 60 + 113.10 = \mathbf{173.10 \text{ mm}}$$

Inside length of one link = $30 + 2 \times 15 = 60 \text{ mm}$. In a chain each link advances the chain by its inside length, so pitch = 60 mm.

$$\text{Number of links} = \frac{60,000}{60} = \mathbf{1,000 \text{ links}}$$

$$\text{Total length of bar} = 1,000 \times 173.10 \text{ mm} = \mathbf{173.10 \text{ m}}$$

$$\text{Volume per link} = \frac{\pi}{4} \times (0.6)^2 \times 17.31 = 0.2827 \times 17.31 = 4.894 \text{ cm}^3$$

$$\text{Weight of chain} = 1,000 \times 4.894 \times 7.7 = 37,685 \text{ g} = \mathbf{37.69 \text{ kg}}$$

Losses: assume 5% for cutting, scale and welding of links.

$$\text{Bar required} = 1.05 \times 173.10 = 181.75 \text{ m; weight} = 1.05 \times 37.69 = 39.57 \text{ kg}$$

14 (b) Elaborate the unique considerations involved in estimating jobs for a foundry shop compared to a welding shop.

Estimating in a foundry shop

1. **Pattern cost** – pattern material and making; spread over the number of castings.
2. **Material cost** – net casting weight + gates, risers and runners (20–40%) + melting loss (3–7%) – credit for returned scrap.
3. **Moulding and core-making cost** – sand, binders, moulding time, cores.
4. **Melting cost** – fuel / electric power per kg of metal.
5. **Fettling and cleaning** – removal of gates, risers, sand; shot blasting.
6. **Inspection and rejection allowance** – defective castings (blow holes, shrinkage) add cost.
7. **Overheads.**

Estimating in a welding shop

1. Weld metal volume from joint geometry; electrode cost.
2. Arc time, operator factor and labour cost.
3. Power / gas cost.
4. Edge preparation, fixtures, overheads.

Comparison

Point	Foundry shop	Welding shop
Main cost driver	Weight of casting and pattern	Length and section of weld
Tooling	Pattern, core boxes – high one-time cost	Fixtures – low cost
Material losses	Gates, risers, melting loss	Stub ends, spatter, slag
Energy	Melting furnace (per kg)	Arc power / gas (per metre)
Quantity effect	Cost falls sharply with batch size	Almost constant per metre
Rejection risk	High (casting defects)	Lower (repairable)

15 (a) Enumerate the steps involved in calculating the machining time for lathe operations with the relevant formulae.

General steps

1. Find the **length of cut** $L = \text{length of job} + \text{approach} + \text{over-travel}$.
2. Find the **spindle speed**: $N = \frac{1000 V}{\pi D}$ rpm.
3. Find the **number of cuts** $= \frac{D - d}{2 \times \text{depth of cut}}$.
4. **Time** $T = \frac{L}{f \times N} \times \text{number of cuts}$.
5. Add set-up, handling and tear-down times for the total time.

Operation-wise formulae

Operation	Length of cut L	Time
Turning	$l + \text{approach} + \text{over-travel}$	$T = L \times \text{cuts} \div (f N)$
Facing	$D/2 (+ \text{approach}); N \text{ at } D \text{ or mean diameter}$	$T = (D/2) \div (f N) \text{ per cut}$
Drilling	$l + 0.3 d$	$T = (l + 0.3 d) \div (f N)$
Chamfering	chamfer length (axial)	$T = L \div (f N)$
Thread cutting	thread length; feed = pitch	$T = L \times \text{number of cuts} \div (p N)$
Knurling	knurl length	$T = L \div (f N)$

Number of thread cuts = depth of thread ($0.613 p$ for metric) $\div$ depth per cut.

15 (b) Shaper: plate 600 mm $\times$ 1200 mm, $V = 15$ m/min, return : cutting = 2 : 3, clearance 25 mm each end on length and 15 mm on width; roughing feed 2 mm/stroke, finishing feed 1 mm/stroke.

$K = \text{return time} \div \text{cutting time} = 2/3$. The stroke is taken along the 1200 mm length.

Length of stroke $L = 1200 + 2 \times 25 = 1250$ mm Width to be covered $B = 600 + 2 \times 15 = 630$ mm

$$\text{Time per stroke} = \frac{L (1 + K)}{1000 V} = \frac{1250 \times (1 + 2/3)}{1000 \times 15} = 0.1389 \text{ min}$$

$$\text{Roughing: strokes} = \frac{630}{2} = 315; \text{ time} = 315 \times 0.1389 = 43.75 \text{ min}$$

$$\text{Finishing: strokes} = \frac{630}{1} = 630; \text{ time} = 630 \times 0.1389 = 87.50 \text{ min}$$

Total machining time = $43.75 + 87.50 = 131.25$ min $\approx$ 2 h 11 min

PART C – 15 MARK ANSWERS

16 (a) Machining time for a stepped pin from 25 mm bar: head 25 mm × 8 mm, step 15 mm × 20 mm, end 10 mm × 30 mm with 20 mm threaded M10 × 1.5. Turning V = 40 m/min, f = 0.4 mm/rev, depth 2.5 mm/pass; threading V = 8 m/min, 0.15 mm per pass.

Approach and over-travel are neglected.

Step 1: Turning Ø25 → Ø15 (length 20 + 30 = 50 mm)

$$\text{Number of cuts} = \frac{25 - 15}{2 \times 2.5} = 2 \quad N = \frac{1000 \times 40}{\pi \times 25} = 509.3 \text{ rpm}$$

$$T_1 = \frac{50}{0.4 \times 509.3} \times 2 = 0.491 \text{ min}$$

Step 2: Turning Ø15 → Ø10 (length 30 mm)

$$\text{Number of cuts} = \frac{15 - 10}{2 \times 2.5} = 1 \quad N = \frac{1000 \times 40}{\pi \times 15} = 848.8 \text{ rpm}$$

$$T_2 = \frac{30}{0.4 \times 848.8} = 0.088 \text{ min}$$

Step 3: Thread cutting M10 × 1.5 (length 20 mm)

$$\text{Depth of thread} = 0.613 \times 1.5 = 0.92 \text{ mm} \quad \text{Number of cuts} = \frac{0.92}{0.15} = 6.13 \approx 7$$

$$N = \frac{1000 \times 8}{\pi \times 10} = 254.6 \text{ rpm} \quad T_3 = \frac{20 \times 7}{1.5 \times 254.6} = 0.367 \text{ min}$$

Operation	Time (min)
Turning Ø25 → Ø15	0.491
Turning Ø15 → Ø10	0.088
Thread cutting	0.367
Total machining time	0.946

Total machining time ≈ 0.95 min per piece (excluding set-up, handling and return strokes)
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16 (b) Evaluate the challenges and limitations of machining time calculation and propose innovative solutions for overcoming them.

Introduction

Machining time is calculated from cutting speed, feed, depth of cut and length of cut. The calculated time is the ideal time; the actual time on the shop floor often differs.

Challenges and limitations

1. **Variation in material** – hardness and machinability vary from batch to batch, so the chosen speed may not be correct.

2. **Tool wear** – as the tool wears, speeds are reduced and tool changes add time not covered by formulae.
3. **Machine condition** – older machines cannot run at recommended speeds and feeds; available spindle speeds are stepped.
4. **Operator skill** – handling, set-up and measuring time vary from worker to worker.
5. **Complex geometry** – contours, tapers and interrupted cuts are hard to model by simple formulae.
6. **Ignored elements** – approach, over-travel, idle return strokes, tool positioning and inspection are often left out.
7. **Allowances are guesses** – fatigue and delay allowances are fixed percentages, not measured values.

Innovative solutions

1. **CAM / CNC simulation** – software computes the exact tool path and cycle time, including rapid moves.
2. **Machining databases** – standard data for speeds and feeds by material and tool grade.
3. **IoT and machine monitoring** – sensors record actual cycle times, which are fed back to correct estimates.
4. **Adaptive control** – the machine adjusts feed in real time for tool wear and material variation.
5. **Time study and MTM** – measured standard times for handling and set-up.
6. **Tool life management** – Taylor's equation ($V T^n = C$) used to plan tool changes.
7. **AI / machine learning** – models trained on past jobs predict time for new jobs.

Conclusion

Formula-based time is a good starting point, but combining it with CAM simulation, shop-floor data and standard time databases gives accurate, reliable machining times for costing and scheduling.